

Lecture 07

Euler Lagrange Equations-II Applications

Euler-Lagrange equation

In the last class, we learnt about the Euler-Lagrange equation, and derived it.

If the function $y(x)$ makes the integral functional

$$S = \int_{x_1}^{x_2} f[y(x), y'(x), x] dx$$

stationary, then $y(x)$ must satisfy the Euler-Lagrange differential equation

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'} = 0$$

Applications of Euler-Lagrange equation

Example 1:

Show that the shortest distance between two points on a plane is a straight line.

Solution:

The element of length on a plane is

$$dS = \sqrt{dx^2 + dy^2} \quad (1)$$

from which the total length is

$$S = \int_{(x_1, y_1)}^{(x_2, y_2)} \sqrt{dx^2 + dy^2} = \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \quad (2)$$

If S is to be minimum, f is identified as

$$f = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \quad (3)$$

Then, the Euler equation becomes

$$\frac{d}{dx} \frac{d}{dy'} \left[\sqrt{1 + y'^2} \right] = 0 \quad (4)$$

where, $y' = \frac{dy}{dx}$. (4) becomes:

$$\frac{d}{dx} \left[\frac{y'}{\sqrt{1 + y'^2}} \right] = 0 \quad (5)$$

$$\frac{d}{dx} \left[\frac{y'}{\sqrt{1+y'^2}} \right] = 0 \quad (5)$$

or,

$$\frac{y'}{\sqrt{1+y'^2}} = \text{constant} \equiv C \quad (6)$$

from which we have

$$y' = \sqrt{\frac{C^2}{1-C^2}} = \text{constant} \equiv a \quad (7)$$

Then,

$$\boxed{y = ax + b} \quad (8)$$

This is the equation of a straight line.

Applications of Euler-Lagrange equation

Example 2:

Consider a particle moving in a constant force field starting at rest from some point (x_1, y_1) to some lower point (x_2, y_2) . Find the path that allows the particle to accomplish the transit in the least possible time.

Solution:

Let us choose the co-ordinate system such that (x_1, y_1) is at the origin. Let the force $\mathbf{F}$ be directed along the positive x-axis as shown in Fig.1.

The time to travel from (x_1, y_1) to (x_2, y_2) is

$$t = \int_{(x_1, y_1)}^{(x_2, y_2)} \frac{ds}{v}$$

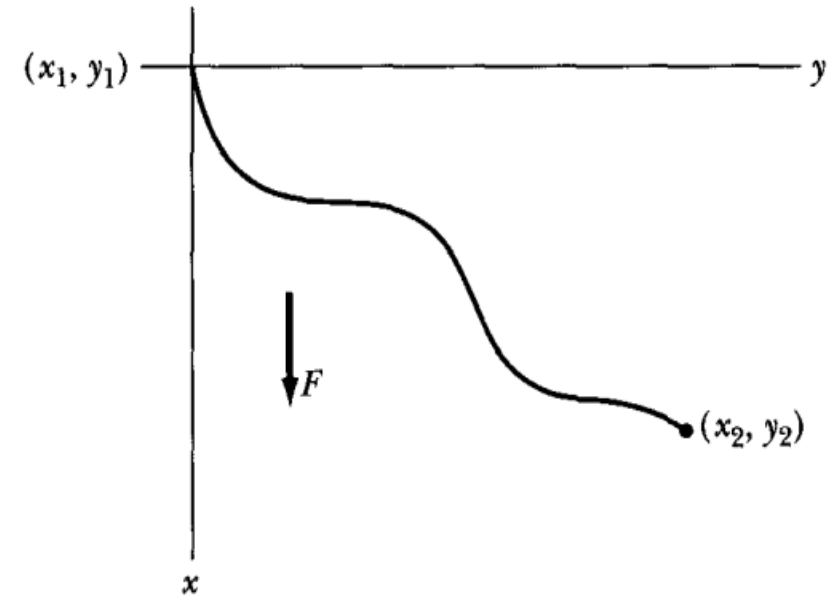


Fig. 1

The force F on the particle is constant. Ignoring the possibility of friction, it is clear that the force is conservative. So, total energy of the particle **$T+V=\text{constant}$** . Here, T refers to the kinetic energy, and V refers to the potential energy.

Let us measure the potential from the point $x=0$, so $V(x=0)=0$. As the particle starts from rest, $T+V=0$.

The kinetic energy is $T = \frac{1}{2}mv^2$

The potential energy is $V = -Fx = -mgx$

Thus,

$$v = \sqrt{2gx}$$

The time required for the particle to make the transit from the origin to (x_2, y_2) is:

$$t = \int_{(x_1, y_1)}^{(x_2, y_2)} \frac{ds}{v} = \int \frac{(dx^2 + dy^2)^{1/2}}{(2gx)^{1/2}}$$
$$= \int_{x_1=0}^{x_2} \left(\frac{1 + y'^2}{2gx} \right)^{1/2} dx \quad ; \quad y' = \frac{dy}{dx}$$

We are looking for the minimum time, so taking cue from the Euler-Lagrange equation, we choose:

$$f = \left(\frac{1 + y'^2}{x} \right)^{1/2} ; \quad \left(\begin{array}{l} \text{The factor } \left(\frac{1}{2g} \right)^{1/2} \\ \text{being a constant} \\ \text{does not affect} \\ \text{the final solution} \end{array} \right)$$

Now let us use the Euler-Lagrange equation:

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0$$

Because, $\frac{\partial f}{\partial y} = 0$ Here, the equation becomes:

$$\frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0$$

$$\frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0$$

$$\Rightarrow \frac{\partial f}{\partial y'} = \text{constant}$$

$$\Rightarrow \frac{\partial}{\partial y'} \left(\frac{1 + y'^2}{x} \right)^{1/2} = \text{constant}$$

$$\Rightarrow \frac{y'}{\{x(1 + y'^2)\}^{1/2}} = \text{constant}$$

$$\frac{y'}{\{x(1+y'^2)\}^{1/2}} = \text{constant}$$

Squaring both sides we get:

$$\frac{y'^2}{x(1+y'^2)} = \text{constant} = \frac{1}{2a} \text{ (say)}$$

The constant is chosen that way for future convenience, as you will see.

So, we have:

$$\frac{y'^2}{x(1+y'^2)} = \frac{1}{2a}$$

$$\Rightarrow y' = \sqrt{\frac{x}{2a-x}}$$

$$\Rightarrow y = \int \sqrt{\frac{x}{2a-x}} dx$$

To solve the integral, let us make the substitution:

$$\boxed{x = a(1 - \cos \theta)} \rightarrow (1)$$

$$dx = a \sin \theta d\theta$$

Hence,

$$y = \int a(1 - \cos \theta) d\theta$$

$$\boxed{y = a(\theta - \sin \theta) + \text{constant}} \rightarrow (2)$$

$$x = a(1 - \cos \theta) \rightarrow (1)$$

$$y = a(\theta - \sin \theta) + \text{constant} \rightarrow (2)$$

We have chosen the initial point to have $x = 0$, $y = 0$; so clearly the initial value of θ is zero. This in turn implies that the constant of integration in (2) is zero.

Thus, the final parametric equation for the path is:

$$\begin{aligned}x &= a(1 - \cos \theta) \\ y &= a(\theta - \sin \theta)\end{aligned}$$

These are the parametric equations for
a **cycloid** passing through the origin.

The constant 'a' is chosen such that the trajectory passes through the specified point (x_2, y_2)

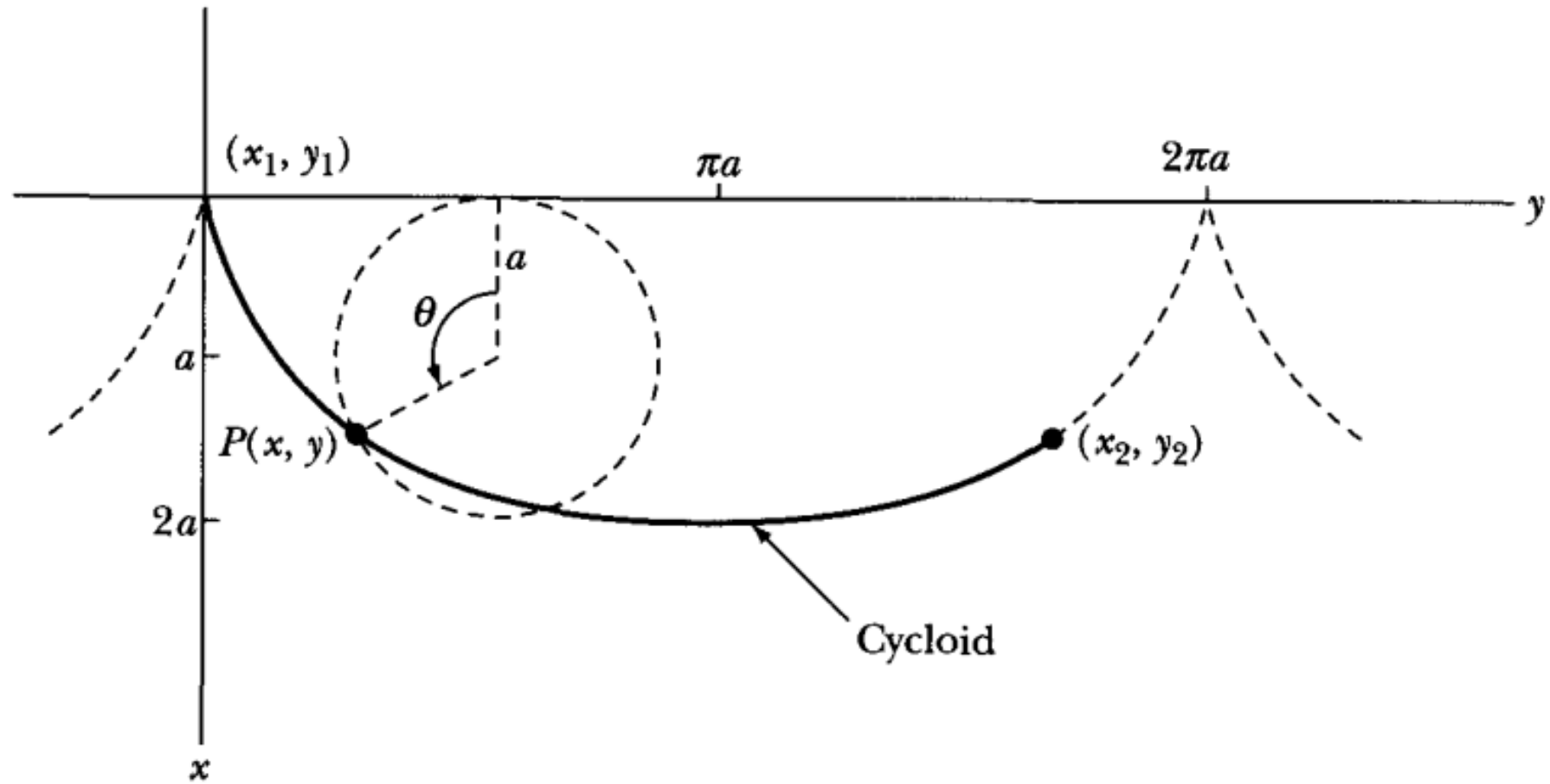


Fig. The trajectory of the particle moving in a constant force field starting at rest from some point (x_1, y_1) to some lower point (x_2, y_2) in the least possible time. **It's a cycloid**

Euler-Lagrange equation with several dependent variables

- Till now, we have considered only problems with just two variables, the independent variable (usually x) and the dependent (usually y).
- The Euler-Lagrange equation we derived earlier is the solution of the variational problem in which it is desired to find the single function $y(x)$ such that the integral of the function f was an extremum.
- In most cases in mechanics, f is a functional of several dependent variables:

$$f = f[y_1(x), y_1'(x), y_2(x), y_2'(x), \dots; x]$$

Or simply, $f = f[y_i(x), y_i'(x); x], i=1,2,\dots,n$

- Exactly the same way, as we did previously, we write:

$$Y_i(x) = y_i(x) + \alpha \eta_i(x)$$

This results in:

$$\frac{\partial S}{\partial \alpha} = \int_{x_1}^{x_2} \sum_i \left(\frac{\partial f}{\partial y_i} - \frac{d}{dx} \frac{\partial f}{\partial y'_i} \right) \eta_i(x) dx$$

Because the individual variations- the $\eta_i(x)$ are all independent,

$\frac{\partial S}{\partial \alpha} = 0$ requires the separate vanishing of each expression in the brackets.

$$\frac{\partial f}{\partial y_i} - \frac{d}{dx} \frac{\partial f}{\partial y'_i} = 0, \quad i = 1, 2, \dots, n$$

Example 3:

Show that the shortest distance between two points in (three-dimensional) space is a straight line.

Solution:

The element of distance in three-dimensional space is

$$dS = \sqrt{dx^2 + dy^2 + dz^2} \quad (1)$$

Suppose x, y, z depends on the parameter t and that the end points are expressed by $(x_1(t_1), y_1(t_1), z_1(t_1)), (x_2(t_2), y_2(t_2), z_2(t_2))$. Then the total distance is

$$S = \int_{t_1}^{t_2} \sqrt{\left[\frac{dx}{dt}\right]^2 + \left[\frac{dy}{dt}\right]^2 + \left[\frac{dz}{dt}\right]^2} dt \quad (2)$$

The function f is identified as

$$f = \sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2} \quad (3)$$

$$f = \sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}$$

Since $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = \frac{\partial f}{\partial z} = 0$, the Euler equations become

$$\left. \begin{aligned} \frac{d}{dt} \frac{\partial f}{\partial \dot{x}} &= 0 \\ \frac{d}{dt} \frac{\partial f}{\partial \dot{y}} &= 0 \\ \frac{d}{dt} \frac{\partial f}{\partial \dot{z}} &= 0 \end{aligned} \right] \quad (4)$$

Thus, we have:

$$\left. \begin{aligned} \frac{\dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}} &= \text{constant} \equiv C_1 \\ \frac{\dot{y}}{\sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}} &= \text{constant} \equiv C_2 \\ \frac{\dot{z}}{\sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}} &= \text{constant} \equiv C_3 \end{aligned} \right] \quad (5)$$

From the combination of these equations, we have

$$\left. \begin{aligned} \frac{\dot{x}}{C_1} &= \frac{\dot{y}}{C_2} \\ \frac{\dot{y}}{C_2} &= \frac{\dot{z}}{C_3} \end{aligned} \right] \quad (6)$$

If we integrate (6) from t_1 to the arbitrary t , we have

$$\left[\begin{array}{l} \frac{x - x_1}{C_1} = \frac{y - y_1}{C_2} \\ \frac{y - y_1}{C_2} = \frac{z - z_1}{C_3} \end{array} \right] \quad (7)$$

On the other hand, the integration of (6) from t_1 to t_2 gives

$$\left[\begin{array}{l} \frac{x_2 - x_1}{C_1} = \frac{y_2 - y_1}{C_2} \\ \frac{y_2 - y_1}{C_2} = \frac{z_2 - z_1}{C_3} \end{array} \right] \quad (8)$$

From (8), we can find the constants.

From (7) and (8) we can obtain:

$$\boxed{\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}}$$

This is the equation expressing a straight line in three-dimensional space passing through the two points (x_1, y_1, z_1) , (x_2, y_2, z_2) .

